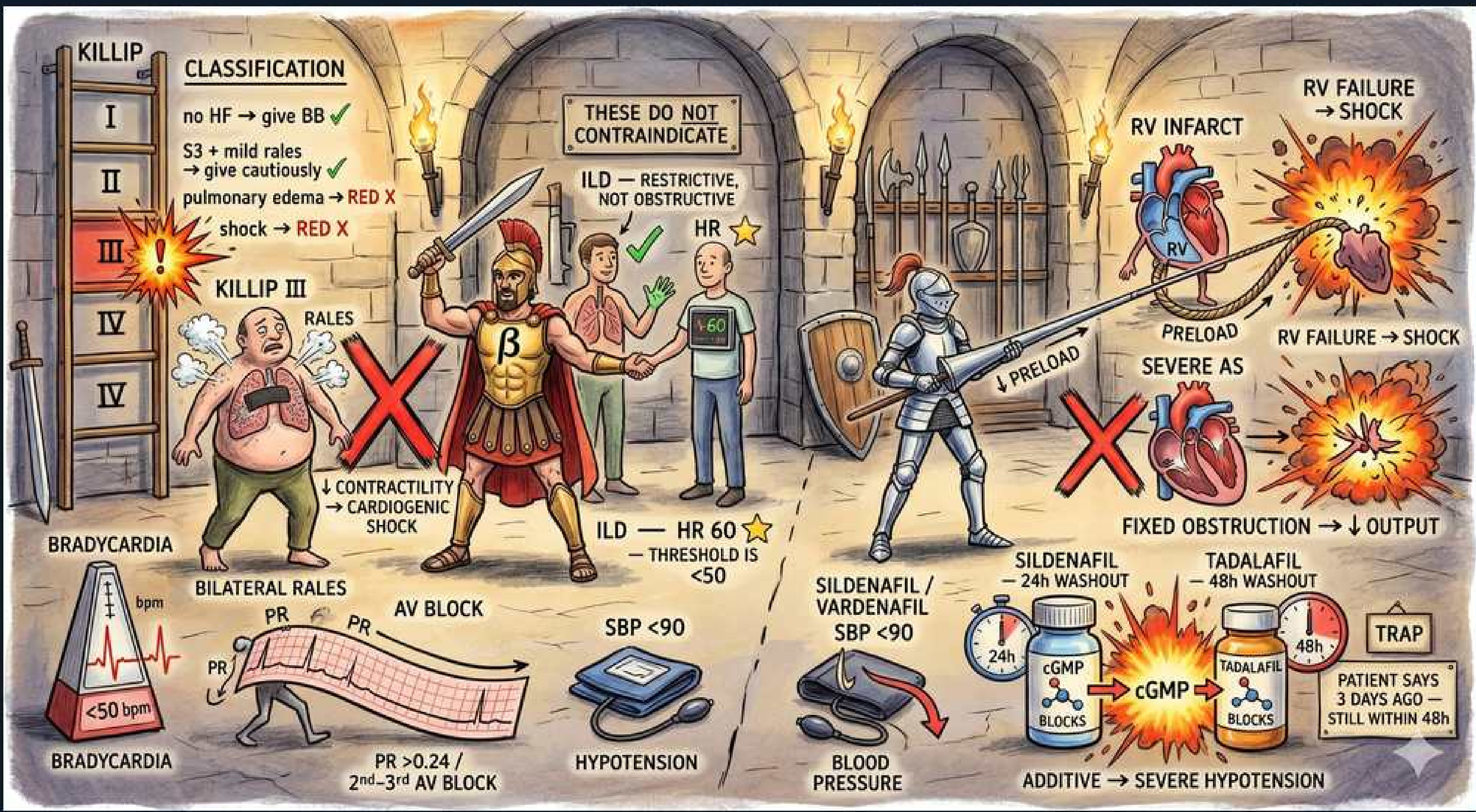
IHD / ACS — NSTE-ACS Diagnosis, Killip & Drug Cautions



1. Killip classification

WHAT YOU SEE

KILLIP CLASSIFICATION ladder I–IV

WHAT IT ENCODES

The Killip class grades acute heart-failure severity at presentation in ACS and is a powerful independent mortality predictor: Class I = no signs of HF (~5% mortality); Class II = rales over <50% of lung fields, S3 gallop, or elevated JVP (~15–20%); Class III = frank pulmonary edema with rales >50% (~30–40%); Class IV = cardiogenic shock with SBP <90 and hypoperfusion (~60–80%). Mnemonic: 'I'm Sick, Soggy, Shocked' for classes I→IV. Each ascending class roughly doubles short-term mortality and shifts management toward urgent invasive strategy and hemodynamic support.

BOARD TRAP

WRONG answer: choosing GRACE or TIMI score as the bedside tool to grade pump failure. Discriminator — Killip is a clinical exam-based HF severity scale specific to ACS; GRACE/TIMI are composite risk calculators (age, vitals, troponin, ECG) for ischemic risk and timing of angiography. Killip class IV (shock) is itself a 'very-high-risk' feature mandating <2h invasive regardless of the GRACE number.

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2. Killip I — give beta-blocker

WHAT YOU SEE

no HF → give BB (green check)

WHAT IT ENCODES

In Killip I (no clinical HF, hemodynamically stable, no contraindication) an ORAL beta-blocker should be started within the first 24h of NSTE-ACS — e.g. metoprolol tartrate 25–50 mg q6–12h or carvedilol 6.25 mg BID, titrated to HR 50–60. Beta-blockade lowers myocardial O2 demand by reducing HR, contractility, and wall stress, shrinks infarct size, and cuts reinfarction and ventricular arrhythmia. Start ORAL, not IV, unless ongoing ischemia/hypertension with no HF.

BOARD TRAP

WRONG answer: giving early IV beta-blockade routinely to everyone. Discriminator — the COMMIT trial showed routine early IV metoprolol increased cardiogenic shock; IV beta-blockers are reserved for persistent hypertension/tachyarrhythmia WITHOUT HF or low-output signs. Even in Killip I, screen for the contraindications (HR <50, SBP <90, PR >0.24, advanced AV block, reactive airway) before dosing.

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3. Killip II — caution (S3, rales)

WHAT YOU SEE

S3 + mild rales → give cautiously (RED X warning)

WHAT IT ENCODES

Killip II = early/mild HF (S3 gallop, bibasilar rales <50% of fields, raised JVP). Beta-blockers may still be used but only LOW-dose, oral, slowly titrated, and after the patient is no longer volume-overloaded/decompensating — because acutely they blunt the compensatory tachycardia and inotropy keeping cardiac output up. Stabilize congestion first (diuresis), then introduce a beta-blocker once euvolemic.

BOARD TRAP

WRONG answer: loading a full-dose beta-blocker immediately because 'beta-blockers are good for heart failure.' Discriminator — beta-blockers improve CHRONIC compensated HFrEF mortality, but in ACUTE decompensation they can tip the patient into low output. Timing matters: chronic HFrEF benefit ≠ acute Killip II benefit.

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4. Killip III/IV — NO beta-blocker

WHAT YOU SEE

pulmonary edema RED X; shock RED X

WHAT IT ENCODES

Killip III (acute pulmonary edema) and Killip IV (cardiogenic shock) are absolute contraindications to beta-blockade in the acute phase — negative inotropy/chronotropy abolishes the catecholamine drive maintaining perfusion and can precipitate fatal hemodynamic collapse. Manage instead with reperfusion/revascularization, diuretics and vasodilators for edema (if BP allows), and inotropes/vasopressors or mechanical support (IABP, Impella) for shock.

BOARD TRAP

WRONG answer: 'beta-blocker to reduce ischemia and slow the heart' in a Killip IV shocked patient. Discriminator — the failing, shock-state heart depends on sympathetic tone; blocking it removes the only thing sustaining output. Defer beta-blockade entirely until the patient is fully compensated, then reintroduce at low dose.

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5. Beta-blocker contraindications

WHAT YOU SEE

Spartan warrior, RED X: ↓contractility / cardiogenic shock, bradycardia, AV block

WHAT IT ENCODES

Core beta-blocker contraindications in ACS: cardiogenic shock or low-output state, decompensated HF (Killip III/IV), bradycardia HR <50–60, high-grade AV block (2nd/3rd degree or PR >0.24 without pacer), SBP <90/hypotension, and active reactive bronchospasm (severe asthma). Cocaine-associated chest pain is a classic relative contraindication (unopposed alpha → coronary vasoconstriction).

BOARD TRAP

WRONG answer: withholding a beta-blocker in a stable COPD patient. Discriminator — COPD and well-controlled asthma are NOT absolute contraindications; cardioselective β1 agents (metoprolol, bisoprolol) are tolerated and reduce mortality. Only severe active bronchospasm truly contraindicates. The hard 'no' list is shock, brady, high-grade block, hypotension.

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6. Bradycardia / HR threshold

WHAT YOU SEE

metronome <50 bpm, BRADYCARDIA

WHAT IT ENCODES

A resting HR <50–60 bpm is a contraindication to starting or up-titrating any AV-nodal blocker (beta-blocker, non-dihydropyridine CCB, ivabradine, digoxin) because further rate suppression risks symptomatic bradycardia or asystole. In ACS the target with beta-blockade is HR 50–60 at rest; if the patient is already there or below from disease or other drugs, hold the agent.

BOARD TRAP

WRONG answer: reflexively withholding the beta-blocker whenever HR sits in the high 50s–60. Discriminator — HR 60 with stable rhythm is acceptable and even target; the contraindication is HR <50 or symptomatic bradycardia. The scene's HR-60 star explicitly marks the safe threshold versus the <50 metronome that is too slow.

7. ILD = restrictive, not obstructive

WHAT YOU SEE

THESE DO NOT CONTRAINDICATE: ILD = RESTRICTIVE not obstructive; HR 60 star

WHAT IT ENCODES

Interstitial lung disease produces a RESTRICTIVE pattern (↓FVC, ↓TLC, normal/high FEV1/FVC ratio) with NO bronchospastic reactive airway component — so it does NOT contraindicate beta-blockers. The bronchospasm caution applies only to OBSTRUCTIVE reactive airway disease (severe asthma) where β2 blockade triggers bronchoconstriction. Restrictive physiology has no β2-mediated airway tone to worsen.

BOARD TRAP

WRONG answer: 'patient has lung disease, so avoid the beta-blocker.' Discriminator — the relevant question is obstructive vs restrictive. ILD/pulmonary fibrosis (restrictive) = safe; severe active asthma (obstructive, reactive) = caution. Boards bait you with the word 'lung disease' to make you over-generalize.

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8. AV block / PR prolongation

WHAT YOU SEE

ECG: PR >0.24, 2nd–3rd degree AV block

WHAT IT ENCODES

PR interval >0.24s (1st-degree, relative caution) and especially Mobitz II, 2nd-degree advanced, or 3rd-degree (complete) AV block are contraindications to AV-nodal blocking drugs — beta-blockers and non-DHP CCBs (verapamil, diltiazem) further depress conduction. In high-grade block without a functioning pacemaker, these agents can produce complete heart block, asystole, or escape-rhythm failure.

BOARD TRAP

WRONG answer: adding verapamil/diltiazem 'for rate control' in a patient with Mobitz II or complete heart block. Discriminator — non-DHP CCBs are just as dangerous as beta-blockers at the AV node; the safe move is a temporary pacemaker first. Also avoid stacking two AV-nodal blockers (beta-blocker + non-DHP CCB) even with normal conduction — synergistic bradycardia/block.

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9. Hypotension SBP <90

WHAT YOU SEE

blood pressure cuff, SBP <90, HYPOTENSION

WHAT IT ENCODES

SBP <90 mmHg contraindicates negative-inotropic and vasodilator therapy in ACS — beta-blockers (drop output), nitrates (drop preload/afterload), and ACE inhibitors (drop afterload) can all worsen end-organ hypoperfusion. In the hypotensive ACS patient, identify the cause (RV infarct → fluids; LV pump failure/shock → inotropes ± mechanical support; arrhythmia → correct rhythm) rather than giving BP-lowering agents.

BOARD TRAP

WRONG answer: giving sublingual nitroglycerin for chest pain in a patient with SBP 85. Discriminator — nitrates require SBP ≥90–100; in hypotension they cause syncope/collapse, and in an RV infarct they are catastrophic. Treat ongoing ischemic pain with morphine/oxygen as needed and fix the hemodynamics first.

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10. RV infarct → preload dependent

WHAT YOU SEE

RV INFARCT, PRELOAD, RV FAILURE → SHOCK

WHAT IT ENCODES

Right-ventricular infarction (usually proximal RCA, accompanying inferior STEMI) makes cardiac output critically PRELOAD-dependent — the stiff, failing RV needs high filling pressures to push blood across to the LV. Clues: inferior MI + hypotension + clear lungs + elevated JVP (Kussmaul sign); confirm with ST elevation in right-sided lead V4R. Treatment = IV fluid bolus to augment preload, avoid all preload reducers, and consider inotropes (dobutamine) if fluids fail.

BOARD TRAP

WRONG answer: giving nitrates or diuretics to an inferior-MI patient who becomes hypotensive. Discriminator — that hypotension is preload failure from RV involvement, not volume overload; nitrates/diuretics drop preload further and cause profound shock. The hypotension-after-nitro response is itself a classic clue to underlying RV infarct.

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11. Severe AS — fixed obstruction

WHAT YOU SEE

SEVERE AS, FIXED OBSTRUCTION → ↓OUTPUT, RED X

WHAT IT ENCODES

In severe aortic stenosis the valve is a FIXED obstruction — the LV cannot raise stroke volume to compensate for a drop in systemic vascular resistance. Nitrates and other vasodilators lower preload/afterload, and because output is fixed, mean arterial pressure falls steeply → syncope, hypotension, and coronary hypoperfusion of an already hypertrophied LV. The hallmark exam is a crescendo-decrescendo systolic murmur radiating to carotids with a slow-rising pulse (pulsus parvus et tardus).

BOARD TRAP

WRONG answer: nitroglycerin for the exertional chest pain of an undiagnosed severe-AS patient. Discriminator — AS angina mimics CAD angina, but a vasodilator across a fixed obstruction causes catastrophic hypotension/syncope. Severe AS (also HOCM and RV infarct) are the classic 'preload/afterload-sensitive, no-nitrate' lesions.

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12. Nitrate + PDE5 inhibitor washout

WHAT YOU SEE

Sildenafil 24h washout / Tadalafil 48h washout; additive → severe hypotension

WHAT IT ENCODES

Nitrates donate NO → ↑cGMP → vasodilation; PDE5 inhibitors block cGMP breakdown → markedly amplified, prolonged ↑cGMP when combined → severe refractory hypotension. Absolute contraindication within the washout window: sildenafil and vardenafil need ≥24h, tadalafil (long-acting) needs ≥48h. If nitrate-induced hypotension occurs, treat with IV fluids and an alpha-agonist (phenylephrine); the offending agent is NOT reversible by stopping the nitrate alone.

BOARD TRAP

WRONG answer: 'he took the ED pill three days ago, so nitrates are safe.' Discriminator — for tadalafil the window is 48h, so 3 days is clear, but a patient who took it ~36h ago is STILL within the tadalafil window despite >24h elapsing. Always pin down the exact drug and exact timing, not just 'a couple days.'
